SIPDEP
DFIR / DEV

Windows Event Log Analysis 101

How Do You Investigate Windows Event Logs?

This sheet highlights key event IDs for Windows event log investigations. Define your scope and goals, review related logs, form a hypothesis, and validate it against other evidence.

Logon

Investigate suspicious logon activity, including the accounts used and their source systems.

Security

4624	Successful logon
4625	Failed logon
4634	Logoff completed
4647	User initiated logoff
4648	Logon attempt using explicit credentials
4672	Special privileges assigned to a new logon
4673	Privileged service called
4674	Attempted operation on a privileged object

Notes: Match 4624 and 4672 by Logon ID on the same host. For 4625, check Status / SubStatus. Event 4648 is logged where explicit credentials are used; confirm authentication results in related logs.

LogonType in 4624 / 4625

2	Interactive logon
3	Network logon
4	Batch logon
5	Service logon
7	Unlock
8	NetworkCleartext
9	Separate credentials for outbound connections
10	Remote interactive logon
11	Cached interactive logon

Caution: Type 3 also occurs in routine activity, such as share access. Type 9 keeps the local identity but uses separate outbound credentials. Type 11 uses cached credentials without querying a DC.

Kerberos / NTLM

Investigate Kerberos ticket abuse and NTLM usage.

Security

4768	Kerberos TGT requested
4769	Service ticket requested
4770	Service ticket renewed
4771	Kerberos preauthentication failed
4776	NTLM credential validation

Notes: For domain accounts, Kerberos authentication and NTLM validation are mainly logged on DCs. Local account NTLM validation is logged on the authenticating endpoint. Event 4776 has no destination host information.

Microsoft-Windows-NTLM/Operational

4020	Outbound NTLM authentication (Information)
4021	Outbound NTLM authentication (Warning)
4022	Inbound NTLM authentication (Information)
4023	Inbound NTLM authentication (Warning)
4030	Cross-domain authentication request (Information)
4031	Cross-domain authentication request (Warning)
4032	Same-domain authentication request (Information)
4033	Same-domain authentication request (Warning)

Notes: Events 4020–4021 are logged on the source, 4022–4023 on the destination, and 4030–4033 on DCs. Availability depends on OS version, installed updates, and audit settings.

Caution: Warning events may indicate weaker NTLM security; they do not necessarily mean authentication failed.

RDP

Trace RDP authentication, connections, disconnects, and reconnects.

Microsoft-Windows-TerminalServices-RemoteConnectionManager/Operational

261	RDP connection received
1146	Session initialization succeeded
1147	Session connection succeeded
1148	Session connection failed
1149	User authentication succeeded

Microsoft-Windows-TerminalServices-LocalSessionManager/Operational

21	Session logon
22	Shell started
23	Session logoff
24	Session disconnected
25	Session reconnected
39	Disconnected by another session
40	Disconnection reason

Security

4778	Session reconnected
4779	Session disconnected

Microsoft-Windows-TerminalServices-RDPClient/Operational

1024	RDP connection attempt
1026	RDP disconnected

Notes: RDPClient logs on the source; the others log on the destination. Event 1149 alone does not establish a successful RDP logon. Correlate with 4624 (Type 10) and LocalSessionManager 21 / 22.

Processes

Investigate suspicious tools, their execution paths, and the accounts that launched them.

Security

4688	Process created
4689	Process terminated

Requirements: Enable process creation auditing and command line logging separately. Check NewProcessName, parent process, CommandLine, and Logon ID. Process IDs can be reused.

PowerShell

Investigate suspicious commands and scripts.

Windows PowerShell

400	Engine started
403	Engine stopped

Microsoft-Windows-PowerShell/Operational

4103	Module logging
4104	Script block logging
4105	Script block execution started
4106	Script block execution completed

Notes: Check HostApplication, ScriptBlockText, and Path; details depend on settings. For split 4104 events, MessageNumber is the sequence number and MessageTotal the total count.

SMB

Investigate lateral movement through administrative shares, as well as file staging and potential data exfiltration.

Security

5140	Network share accessed
5142	Share added
5143	Share modified
5144	Share deleted
5145	Access check for an object within a share

Notes: Check the share name (`ADMIN$`, `C$`, `IPC$`), target path, AccessMask, and Logon ID. Event 5145 records an access check, not the transfer or exfiltration itself.

Microsoft-Windows-SMBServer/Security

551	SMB session authentication failed
1006	Share access denied
1009	Anonymous access denied

Microsoft-Windows-SMBClient/Connectivity

30800	Name resolution failed
30803	Network connection failed
30805	Session lost
30806	Session reestablished
30807	Share connection lost
30809	Timed out waiting for server response
31001	Security context initialization failed
31010	Connection to share failed

Notes: Connection errors also occur during network outages and routine operations.

Services

Investigate remote execution, persistence, and unexpected stops of security or backup services.

Security

4697	Service installed

System

7036	Service started / stopped
7040	Service start type changed
7045	Service installed

Notes: Check the executable path, service account, and start type. The System provider is Service Control Manager. Legitimate software installation and updates also generate these events.

WMI

Investigate remote WMI activity and persistence through permanent event subscriptions.

Microsoft-Windows-WMI-Activity/Operational

5857	WMI provider started
5858	WMI operation error
5859	Event subscription activity
5861	Permanent consumer / binding activity

Notes: Check ClientProcessId, Namespace, Filter, and Consumer. Routine administration also generates these events.

Task Scheduler

Investigate delayed or recurring execution and persistence via scheduled tasks.

Security

4698	Task created
4699	Task deleted
4700	Task enabled
4701	Task disabled
4702	Task updated

Microsoft-Windows-TaskScheduler/Operational

100	Task started
101	Task failed to start
102	Task completed
103	Task instance failed to start
106	Task registered
107	Triggered by a schedule
108	Triggered by an event
110	Triggered by a user
118	Triggered by OS startup
119	Triggered by logon
129	Task process created
140	Task registration updated
141	Task registration deleted
142	Task disabled
200	Action started
201	Action completed
203	Action failed to launch

Notes: Check TaskName, TaskContent XML, the executing user, arguments, and triggers. Registration and execution are recorded in separate events.

Accounts & Groups

Investigate suspicious account creation and privileged group membership changes.

Security

4720	Account created
4722	Account enabled
4723	Password change attempted
4724	Password reset attempted
4725	Account disabled
4726	Account deleted
4728	Member added to a global group
4729	Member removed from a global group
4732	Member added to a local group
4733	Member removed from a local group
4738	Account changed
4740	Account locked out
4756	Member added to a universal group
4757	Member removed from a universal group

Notes: Subject identifies the actor, Target the affected account or group, and Member the added or removed member. Privileged groups include Administrators and Domain Admins.

AD & Audit Policy

Investigate suspicious AD attribute changes and weakened audit settings.

Security

4719	System audit policy changed
4739	Domain policy changed
5136	AD object modified
5137	AD object created
5139	AD object moved
5141	AD object deleted

Notes: AD changes are mainly recorded on DCs and require audit policy and SACL settings. Identify the object using ObjectDN or related fields. In 5136, check AttributeLDAPDisplayName and AttributeValue for the changed attribute and value.

Firewall

Investigate suspicious destinations and weakened firewall settings.

Security

4946	Firewall rule added
4947	Firewall rule modified
4948	Firewall rule deleted
4949	Firewall settings reset
4950	Firewall setting changed
4954	Firewall Group Policy settings changed / applied
4956	Active profile changed
5152	Packet blocked
5154	Listen allowed
5155	Listen blocked
5156	Connection allowed
5157	Connection blocked
5158	Bind allowed
5159	Bind blocked

Notes: For network activity, check Application, source / destination IPs and ports, and Direction. For rule changes, check RuleId, RuleName, and ProfileChanged. WFP auditing must be enabled and can generate high volumes. Payloads are not recorded.

Microsoft Defender

Investigate malware detections, remediation outcomes, and when protection was disabled.

Microsoft-Windows-Windows Defender/Operational

1116	Malware / PUA detected
1117	Remediation action taken
1118	Remediation failed
1119	Remediation failed with a critical error
5001	Real-time protection disabled
5004	Real-time protection configuration changed
5007	Antivirus setting changed
5010	Malware / PUA scanning disabled
5012	Virus scanning disabled
5013	Change blocked by Tamper Protection

Notes: Check ThreatName, Path, Action, and Result. Review detection and remediation outcomes separately. The channel name varies by environment.

Event Logs

Investigate missing or cleared logs and determine the time span covered by retained events.

Security

1100	Event Log service stopped
1101	Audit events dropped
1102	Security log cleared
1104	Security log full
1105	Event log automatically backed up

System

104	Eventlog: Log cleared

Notes: Disabled auditing, overwrites, logging interruptions, and deletion can cause gaps. Log size and behavior when full are configured per channel.

Preservation: Save the original EVTX files and check the oldest and newest timestamps. Security logs can be overwritten quickly.

Time & Power

Investigate restarts, clock changes, and the sequence of events.

Security

4616	System time changed

System

12	Kernel-General: OS started
13	Kernel-General: OS shutdown started
41	Kernel-Power: Reboot without clean shutdown
1074	User32: Shutdown / restart initiated
6005	EventLog: Service started
6006	EventLog: Service stopped
6008	EventLog: Previous shutdown was unexpected
6009	EventLog: OS information at startup
6013	EventLog: System uptime

Notes: SystemTime in XML is recorded in UTC. Events 41 and 6008 indicate an unclean shutdown. Consult other logs when investigating the cause.

Applications

Investigate crashes of suspicious tools and unexpected termination of security software.

Application

1000	Application Error: Crash
1001	Windows Error Reporting: Error report
1002	Application Hang: Hang

Notes: Check the executable, faulting module, exception code, and timestamp. Each ID above applies to the provider named in its row.

SIPDEP - How Do You Investigate Windows Event Logs?
https://sumeshi.github.io/posts/knowledges/windows-eventlog-analysis-101

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For reference only; accuracy and completeness are not guaranteed.
Logging depends on audit settings and OS configuration. Missing events do not prove an action did not occur.